

Startup Robot

Startup UI:

1. Computer login

- 1.1. User: Yujie
- 1.2. Password: yujie312

2. Sourcing and building ROS2

- 2.1. In terminal CLI, source ROS2:
 - ◆ source /opt/ros/humble/setup.bash
- 2.2. Navigate to ros2_ws directory from home directory
 - ◆ cd robot_ws
- 2.3. Source workspace (build files in ~/ros2_ws/install)
 - ◆ source install/setup.bash

3. Running sawGalilController

- 3.1. Navigate to the json configuration file directory
 - ◆ cd ros2_ws/cisst-saw/sawGalilController/core/share/test/
- 3.2. Start galil_controller and attached json config file. You should see a GUI appear. No need to interact with it.
 - ◆ galil_controller -j NPR4.json
- 3.3. Verify correction ros topics are being published. We should see relevant CRTK rostopics: measured_cp, measured_js, move_cp, move_js, etc. See CRTK library documentation for more info.
 - ◆ ros2 topic list

4. Run the main robot controller GUI script

- 4.1. Repeat step 2 (Sourcing and building ROS2) in a new terminal window.
- 4.2. Run GUI and control script. You should see a GUI appear with controls for the robot.
 - ◆ ros2 run needlePenetrationRobot insertion_experiment

UI User Guide

The UI has 3 tabs:

1. Control
2. Insertion Experiment
3. Manual Homing

1. Control

NPR Control Panel v0.2.0

Position:

X

0.0001

Y

0.0000

Z

0.0000

D

0.0000

Velocity: vx=0.0000, vy=0.0000, vz=0.0000, vd=0.0000

Status: Disabled

Command: Ready

Homing: Not Homed

X: Not Homed

Y: Not Homed

Z: Not Homed

D: Not Homed

Control

Insertion Experiment

Manual Homing

Jog Controls

↑

↓

↑

↓

↑

↓

↑

↓

X

Y

Z

D

Auto-Home (X,Y,Z)

Manual Homing

Disabled

Set Position

X: X Setpoint [-0.05, 0.05]

Y: Y Setpoint [-0.05, 0.05]

Z: Z Setpoint [-0.05, 0.05]

D: D Setpoint [-0.07, 0.065]

Set Position

Console Output:

2. Insertion Experiment

NPR Control Panel v0.2.0

Position:

X

0.0001

Y

0.0000

Z

0.0000

D

0.0000

Velocity: vx=0.0000, vy=0.0000, vz=0.0000, vd=0.0000

Status: Disabled

Command: Ready

Homing: Not Homed

X: Not Homed

Y: Not Homed

Z: Not Homed

D: Not Homed

Control

Insertion Experiment

Manual Homing

Insertion Experiment Instructions:

1. Make sure the robot is enabled and homed

2. Select your needle length

3. Select the y-axis index for insertion

4. Click 'Move to start' to bring robot to start position for experiment

Set Up Insertion

Select needle length

Select index

Move to start position

Perform Insertion

Select insertion type

Insert

Console Output:

3. Manual Homing

NPR Control Panel v0.2.0

Position:

X

0.0001

Y

0.0000

Z

0.0000

D

0.0000

Velocity: vx=0.0000, vy=0.0000, vz=0.0000, vd=0.0000

Status: Disabled

Command: Ready

Homing: Not Homed

X: Not Homed

Y: Not Homed

Z: Not Homed

D: Not Homed

Control

Insertion Experiment

Manual Homing

Manual Homing Instructions:

1. Make sure the robot is enabled

2. Use the jog controls to align each axis with its reference mark

3. Click 'Set Home Position' for that axis

4. Repeat for all axes that need to be manually homed

X Axis

Current Position: 0.0001

Jog Controls:

←

X

→

Home Position Value:

0.0000

Set X Home Position

Y Axis

Current Position: 0.0000

Jog Controls:

←

Y

→

Home Position Value:

0.0000

Set Y Home Position

Z Axis

Current Position: 0.0000

Jog Controls:

←

Z

→

Home Position Value:

0.0000

Set Z Home Position

D Axis

Current Position: 0.0000

Jog Controls:

←

D

→

Home Position Value:

0.0000

Set D Home Position

Reset All Home Positions

Return to Control Panel

Console Output: